

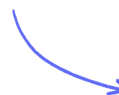
IGCSE · Cambridge (CIE) · Maths

 25 mins  4 questions

Modelling: Road Traffic

Modelling

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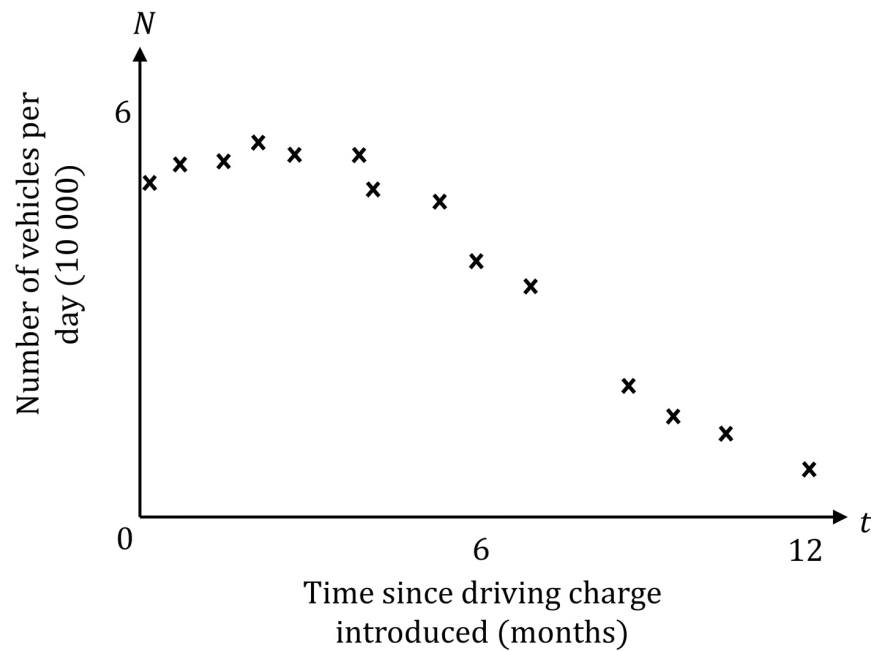
Total Marks

/25

1 (a) This task looks at models for daily road traffic on a circular ring road around a city.

To help reduce the amount of traffic, the mayor of the city has implemented a new daily driving charge on the ring road, which will last for a year.

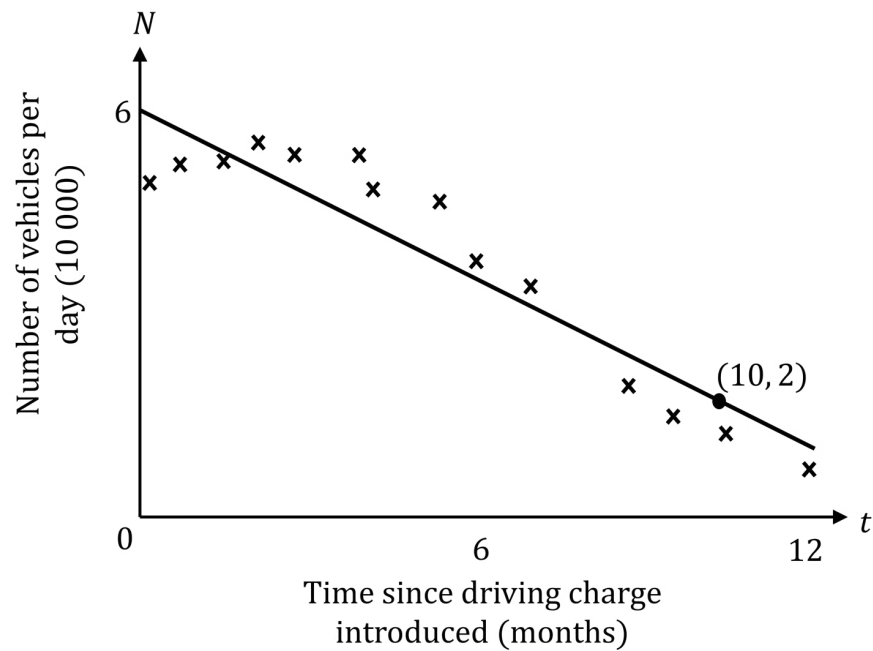
The scatter graph below shows the number of vehicles per day, N , measured in tens of thousands, that use the ring road against the time, t , measured in months, from the date of the driving charge being introduced.



A simple model to relate N and t is given by the line of best fit:

$$N = a + bt$$

The line of best fit intersects the N -axis at the point $N=6$ and passes through the point $(10, 2)$, as shown.



Use this information to find the values of a and b .

(3 marks)

- (b)** Use the model in part (a) to predict the total number of vehicles per day on the ring road 7 months after the driving charge was introduced.

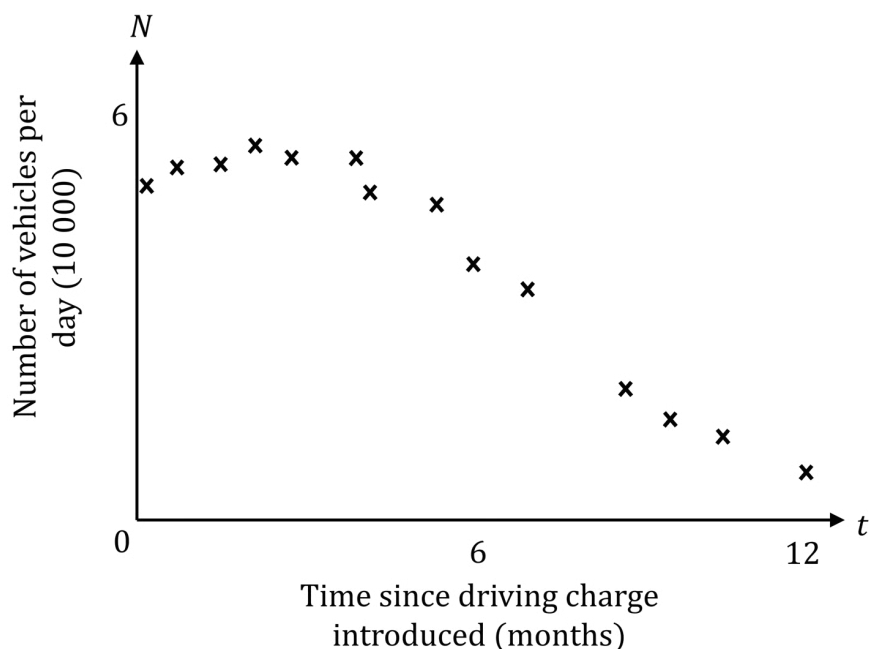
(2 marks)

- (c)** Explain, with reason, whether or not the line of best fit should be continued beyond 12 months.

(1 mark)

- 2 (a)** To help reduce the amount of traffic in a city, the mayor of the city has implemented a new daily driving charge on the ring road, which will last for a year.

The scatter graph below shows the number of vehicles per day, N , measured in tens of thousands, that use the ring road against the time, t , measured in months, from the date of the driving charge being introduced.



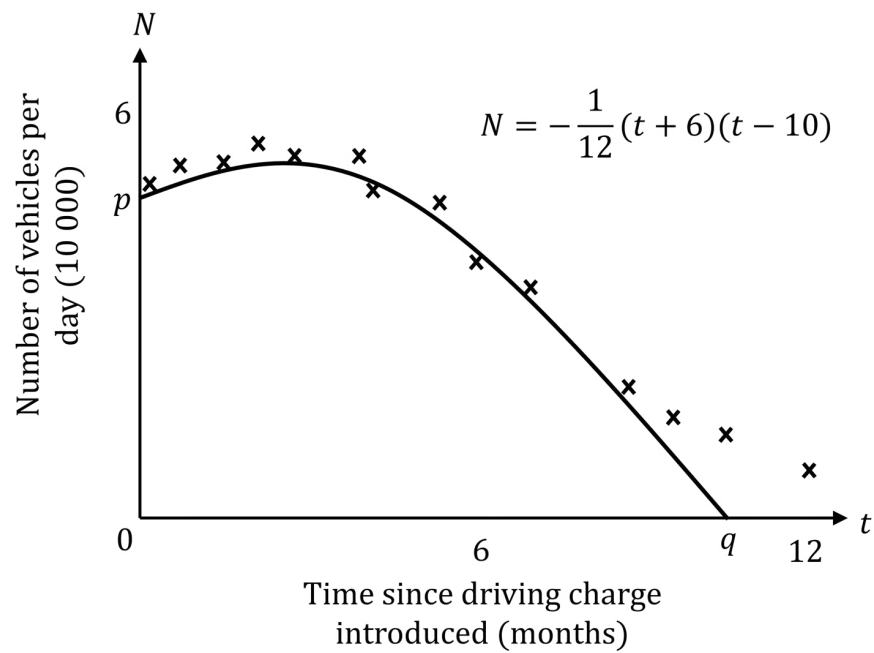
The mayor notices that there is a slight increase in road traffic immediately after the driving charge is introduced, after which the traffic starts to decrease.

The mayor believes this is due to a delay in drivers realising that the new charge had been introduced.

To model this delay, a negative quadratic model is proposed in the form

$$N = -\frac{1}{12}(t+6)(t-10)$$

A sketch of this model is shown below.



Find the values of the axes intercepts, p and q .

(2 marks)

(b) Explain what the value of p represents in relation to the context of the question.

(1 mark)

(c) The maximum point on the model occurs 2 months after the driving charge is introduced.

Find the maximum number of vehicles per day predicted by this model.

Give your answer correct to the nearest hundred.

(1 mark)

(d) The length of the delay is the time it takes, from introducing the new driving charge, for N to first start decreasing below $N = p$.

Find the length of the delay.

You may use any information from part (c) if it helps.

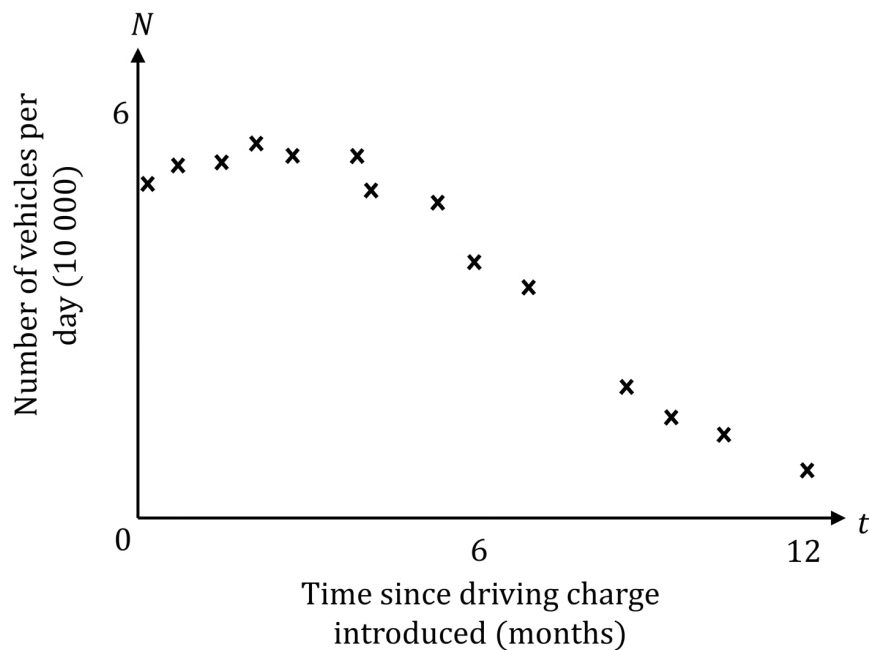
(1 mark)

(e) Explain why the model should not be used from 10 months onwards.

(1 mark)

- 3 (a)** To help reduce the amount of traffic in a city, the mayor of the city has implemented a new daily driving charge on the ring road, which will last for a year.

The scatter graph below shows the number of vehicles per day, N , measured in tens of thousands, that use the ring road against the time, t , measured in months, from the date of the driving charge being introduced.



The mayor suspects that the reduction in traffic will eventually settle to a positive value, rather than decrease to zero.

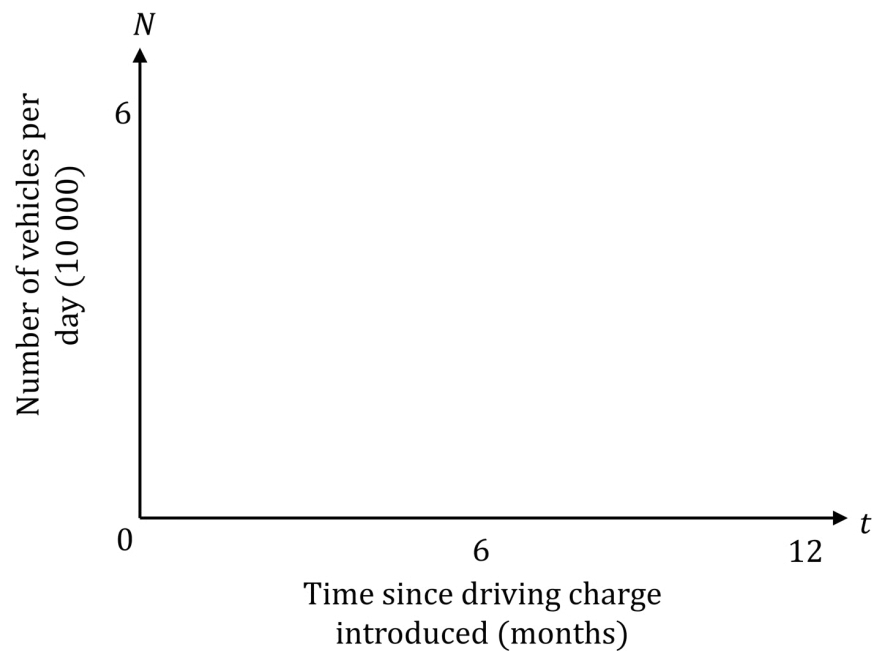
A cubic model is proposed.

A data analyst inputs the data from the scatter graph into a computer to find the closest matching cubic curve.

The computer outputs:

$$N = \frac{1}{108}(t^3 - 21t^2 + 72t + 540)$$

Sketch this model on the axes provided.



(2 marks)

- (b) Find, correct to the nearest hundred, the maximum number of vehicles per day predicted by this model.

(1 mark)

- (c) A negative quadratic model used to model the data predicted a maximum number of vehicles per day of 53300, to the nearest 100 vehicles.

Determine which out of the cubic model or the negative quadratic model predicts the highest maximum number of vehicles per day.

(1 mark)

- (d) The mayor believes the driving charge will have been a success if the number of vehicles per day reduces to below 11 000 by the end of the year.

Does the cubic model predict that the driving charge will have been a success?

Explain your answer clearly.

(2 marks)

(e) The negative quadratic model used was of the form

$$N = -\frac{1}{12}(t+6)(t-10)$$

The mayor wishes to display the negative quadratic model and the cubic model on the same set of axes.

Find the coordinates of the two points of intersection between the negative quadratic model and the cubic model, where $0 \leq t \leq 12$.

(2 marks)

- 4 (a)** To help reduce the amount of traffic in a city, the mayor of the city has implemented a new daily driving charge on the ring road, which will last for a year.

The mayor is also interested in the amount of carbon dioxide emitted by road traffic.

An average vehicle is known to emit around 0.1 kg/km of carbon dioxide.

If the radius of the circular ring road is 8 km, show that an average vehicle would emit around 5.03 kilograms of carbon dioxide when doing one full lap of the ring road.

(2 marks)

- (b)** The mayor assumes that an average vehicle travels **half** a lap of the ring road every day.

The variable c is the total amount of carbon dioxide emitted by **all** vehicles on the ring road per day, measured in kilograms.

Find the value of k in the relationship

$$c = kN$$

where N is the number of vehicles per day, measured in tens of thousands.

Give your value of k correct to 2 significant figures.

(2 marks)

- (c)** A cubic model used to calculate the traffic on the ring road is given by

$$N = \frac{1}{108}(t^3 - 21t^2 + 72t + 540)$$

where N is the number of vehicles per day, measured in tens of thousands, and t is the time, measured in months, from the date of the driving charge being introduced.

The mayor wants to adapt the cubic model into a model that shows:

- the total amount of carbon dioxide emitted by all vehicles on the ring road per day, c , measured in kilograms
- against the time, t , measured in months, from the date of the driving charge being introduced

Use the information in part (b) to write down a cubic model for c in terms of t .

(1 mark)